



Advanced Linear Algebra: Foundations to Frontiers

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1.4.3 The conditioning of an upper triangular matrix



01.4.3 Conditioning of a random triangular matrix

fit width



We now revisit the material from the launch for the semester. We understand that when solving $Lx = b$, even a small relative change to the right-hand side b can amplify into a large relative change in the solution $\hat{x}$ if the condition number of the matrix is large.

Homework 1.4.3.1. Change the script

[Assignments/Week01/matlab/Test_Upper_triangular_solve_100.m](#) to also compute the condition number of matrix U , $\kappa(U)$. Investigate what happens to the condition number as you change the problem size n .

Since in the example the upper triangular matrix is generated to have random values as its entries, chances are that at least one element on its diagonal is small. If that element were zero, then the triangular matrix would be singular. Even if it is not exactly zero, the condition number of U becomes very large if the element is small.